

CSE331 Assignment 2 Part A

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- Deadline for entire Assignment 2 part A submission:
22 August, 2026, 11:59 pm.
 - The submitted document should have clear and legible scans.
1. Prove that the following languages are nonregular using the Pumping lemma:
 - a. $L = \{ w \in \{0, 1\}^* \mid 0^i 1^j 0^k, i + k > j, i, j, k > 0 \}$
 - b. $L = \{ a^n b^l a^k \mid k > n + l \}$
 - c. $L = \{ 0^n l \mid n \geq 0 \}$
 - d. $L = \{ a^n b^l \mid n \neq l \}$
 - e. $L = \{ w \in \{0, 1\}^* \mid w \text{ is a palindrome} \}$
 2. Write CFG for the following languages:
 - a. $L = \{ a^n b^m \mid m, n > 0, n \neq m \}, \Sigma = \{a, b\}$
 - b. $L = \{ a^n b^m \mid m, n > 0, n = 4m + 2 \}, \Sigma = \{a, b\}$
 - c. $L = \{ a^n b^m c^p \mid m, n, p > 0, m=n \text{ or } m=p \}, \Sigma = \{a, b\}$
 - d. $L = \{ w_1 \# w_2 \mid \text{the number of 01 in } w_1 \text{ is equal to the number of 10 in } w_2 \}, \Sigma = \{0, 1\}$
 - e. $L = \{ w_1 \# w_2 \mid w_1 \text{ contains 0 in every 3rd position, } |w_2| = 2|w_1| \}, \Sigma = \{0, 1\}$
 - f. $L = \{ w_1 \# w_2 \mid |w_1| \text{ is odd and } |w_2| \text{ is even, } w_1 \text{ contains the substring 100} \}, \Sigma = \{0, 1\}$
 3. Answer from the following grammar:
$$\begin{aligned} S &\rightarrow PE \\ P &\rightarrow aPa \mid bPb \mid a \mid b \\ E &\rightarrow aaE \mid abE \mid baE \mid bbE \mid \varepsilon \end{aligned}$$
 - a. Give a leftmost derivation for the string **bababba**.
 - b. Sketch the parse tree corresponding to the derivation you gave in (a).
 - c. Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string.
 - d. Find how many parse trees the string **aababab** has in the given grammar.